

CLINICAL INVESTIGATION

Eye

UVEAL MELANOMA RECURRENCE AFTER FRACTIONATED PROTON BEAM THERAPY: COMPARISON OF SURVIVAL IN PATIENTS TREATED WITH REIRRADIATION OR WITH ENUCLEATION

LAURA MARUCCI, M.D.,* MAREK ANCUKIEWICZ, PH.D.,† ANNE MARIE LANE, M.P.H.,‡
JOHN M. COLLIER, PH.D.,† EVANGELOS S. GRAGODAS, M.D.,‡ AND JOHN E. MUNZENRIDER, M.D.†

*Department of Radiation Oncology, Regina Elena Research Institute, Rome, Italy; †Department of Radiation Oncology, Massachusetts General Hospital, Boston, MA; and ‡Retina Service, Massachusetts Eye and Ear Infirmary, Boston, MA

Purpose: To retrospectively compare survival in recurrent uveal melanoma, between patients treated by enucleation or by a second course of fractionated proton beam therapy (PBT).

Methods and Materials: Tumor recurrence was documented in 73 patients treated with PBT for uveal melanoma. Of the patients, 31 received a second course of PBT and 42 underwent enucleation. The mean patient age was 56 and 61 years for those undergoing enucleation and those undergoing reirradiation, respectively. Both primary and recurrent tumors were larger in patients undergoing enucleation. Tumor location and the presence or absence of ciliary body involvement did not differ significantly between the groups. The median follow-up after enucleation and after re-treatment was 79 and 59 months, respectively. Cumulative rates of outcomes and differences in rates between the reirradiated and enucleation groups were calculated by the Cox proportional hazards model and the log-rank test, respectively.

Results: The median survival duration in the enucleated and reirradiated groups was 42 and 90 months, respectively. The median time free of metastases was 38 months in enucleated patients and 97 months in reirradiated patients. At 5 years after enucleation and after reirradiation, the probability of overall survival was 36% and 63%, respectively ($p = 0.040$, log-rank test); the probability of freedom from metastases was 31% and 66%, respectively ($p = 0.028$, log-rank test). These differences persisted after adjustment for recurrent tumor largest diameter and volume at the time of reirradiation or enucleation.

Conclusions: This retrospective analysis suggests that survival in reirradiated patients is not compromised by administration of a second course of PBT for recurrent uveal melanoma. © 2011 Elsevier Inc.

Uveal melanoma, Proton therapy, Recurrent cancer, Re-treatment, Enucleation.

INTRODUCTION

Uveal melanoma is the most common ocular tumor. Radiation therapy has replaced enucleation as a standard treatment for primary uveal melanomas (1). The principal radiation techniques used to treat uveal melanoma are brachytherapy (radioactive plaques) and proton beam therapy (PBT). The choice of treatment often depends on tumor characteristics, that is, tumor size, location (2), and treatment option available to the patient. Proton beam therapy offers the advantage of a more uniform dose delivered to the tumor with a sharp decrease in dose outside the target. These characteristics of proton beams allow the treatment of large tumors with some sparing of the radiosensitive normal structures such as the lens, the optic disc, and the macula from radiation effects.

Given the success of PBT, more facilities have been built worldwide, and an increasing number of articles with large

series of patients have been published (1, 3–8). All report excellent results in terms of local control (95–98%) and overall survival at 5 years (78–82%).

Despite such good local control of primary tumors, 2% to 5% of the patients have recurrence locally and require a second treatment. The most common treatment of recurrent melanoma is enucleation. Since 1984, at the Harvard Cyclotron Laboratory, selected patients with recurrent tumors were offered the possibility of a second treatment with PBT to preserve the eye and possibly some vision.

In a previous publication we reported on the outcome of proton beam reirradiation in 31 patients with recurrent uveal melanoma (9). The results showed a local control rate of 69%, eye retention rate of 55%, and overall survival rate of 64% at 5 years after a second course of PBT. This 5-year survival rate is lower than the 5-year rate (79%) observed after treatment of the primary tumor (1).

Reprint requests to: Marucci Laura, M.D., Department of Radiation Oncology, Regina Elena Institute, Via Chianesi 54, 00144 Rome, Italy. Tel: (+39) 06 52663018; Fax: (+39) 06 52665686; E-mail: marucci_laura@yahoo.com

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